Transit-Node Site Screening — system architecture

A deterministic Python pipeline owns every score. Two Claude-powered components work inside guardrails: an LLM data audit and a bounded follow-up agent.

Deterministic Python · Data store · LLM (Claude) · Plain-code guard · Human in the loop · Output

INPUTS

Open public data — no paid APIs
OpenStreetMap (Overpass) · Overture Maps · Berlin LOR / MSS 2025 · IHK Gewerbedaten
Wikidata + Wikipedia pageviews · VBB GTFS timetable · DB InfraGO station price classes

Site input
store concept + 3 candidate sites (JSON)

Calibration
139 Berlin stations → P10/P90 anchors

Outputs
Diligence ranking · business fit + evidence quality
evidence gaps · field-research checklist

DETERMINISTIC PIPELINE · PYTHON

Ingest
API clients with retries + mirror fallback

Snapshot store
SQLite, append-only
data-as-of ≠ fetched
unknown ≠ 0

Clean & conflate
dedupe · cross-provider POI conflation · opening-hours parser

Metrics
people per day = residents + workers + visitors; competitors

Gap detection
missing · stale · conflicting inputs that could change a call

Scoring
anchored 0–100 scales business fit and evidence quality

Sensitivity
inputs first: joint downside, then weights

Report
sourced Markdown · 42 citations re-checked vs. the DB

stored records · *reviewed corrections* · *gaps*

AI / LLM / AGENT LAYER · ANTHROPIC CLAUDE

① LLM data-plausibility audit

review only; never edits a number

Candidate rules
Python lists likely-wrong records:
whole-company headcount at one address · fame » capacity · missing employer · duplicates

Claude Sonnet 5.5
Messages API · tool use
web search + web fetch
strict-schema findings (report_finding), sources cited

Audit guard
validates every finding
USD cap per run
frozen prompt hash (SHA-256)
required on unseen nodes

Analyst review
accepts / rejects each flag in the review queue (CSV)
audit_apply writes accepted rows only

Correction tables
verified on-site headcounts · unregistered employers · venue attendance · POI exclusions

Evaluated before adoption
answer key + holdout fixed before any code · 4/6 holdout issues caught · 57/58 flags confirmed at unseen stations

② Bounded follow-up agent

decides which gaps get live calls; never sees a score

Gap list
only gaps that could change a conclusion; no raw-count triggers

Pluggable decision policy
- A · Rule baseline (detection order)
- B · Impact-ordered rule — default
- C · Claude Opus 5.5 — tool-use loop

Executor guard
allowed tool only
budget: 4 calls / run, 2 per site
no retry of a failed source
LLM error → falls back to rule

Live tools (refetch)
refetch_osm_pois · refetch_ihk_gewerbedaten
refetch_lor_population

Field flag
flag_field_research + LLM-written field question

Evaluated
16 pre-registered scenarios: A 10/16 · B 14/16 · C 16/16
($0.019 per run)

field checklist

refetch → new snapshot → pipeline reruns

AI guardrails (both components)

AI never writes a score · Plain-code guard between model and data · Human approval for every data change · Hard budgets: calls, web searches, USD · Rule fallback on any API error

ENGINEERING & QUALITY

Python 3.9 / 3.12 · SQLite · 6 tables · Anthropic Python SDK · Claude Opus 5.5 · Sonnet 5.5 · Tool use · web search / fetch · unittest · 241 tests · GitHub Actions CI · Offline reproducibility check · CLI: validate · demo · refresh · Pre-registered AI evaluations · Prompt-leakage test vs. answer key · Mocked LLM client + one real-SDK round trip · Geospatial: haversine, area-weighted catchments · Opening-hours parser (overnight spans)